

Module 1

GraphQL

GraphQL

Definition (GraphQL Foundation):

- GraphQL is a query language for APIs and a runtime for fulfilling those queries with your existing data. It describes your data, lets clients request exactly what they need, and evolves safely over time.
- Key Benefits:
- Reduced network traffic – no over-fetching or under-fetching
- Flexible & language-agnostic – works with any backend
- Simplified data fetching – one request, structured response
- Easy to maintain – schema-based evolution
- Strong typing – catch errors early
- Great dev experience – ideal for mobile apps, public APIs, and microservices

GraphQL

The GraphQL Advantage:

Traditional REST API Call

To load user info + their posts:

1. GET /user/123
2. GET /user/123/posts

- Multiple round-trips
- Over-fetching / under-fetching

GraphQL

```
query {  
  user(id: "123") {  
    name  
    posts {  
      title  
      publishedAt  
    }  
  }  
}
```

- One request
- Only the data needed
- Structured JSON response

GraphQL

GraphQL uses a type system to define your API.

```
type User {  
  id: ID!  
  name: String!  
  email: String!  
  posts: [Post]!  
}
```

```
type Post {  
  title: String!  
  publishedAt: String!  
}
```

- `String!` means it's required
- Relationships (e.g. `posts`) are embedded
- Tools can auto-generate documentation
- Clients get predictable responses
- Catches errors at development time

GraphQL

GraphQL is schema-based, so you can:

- Add new fields:

```
type User {  
  age: Int    # Added later  
}
```

- Deprecate old fields:

```
email: String @deprecated(reason: "Use contactEmail")
```

- No breaking changes
- Versionless API evolution
- Better long-term maintainability

GraphQL with Tyk

Native GraphQL Support

- No middleware required
- Fully compliant with latest GraphQL specs:
 - Queries – Fetching data
 - Mutations – Modifying data
 - Subscriptions – Real-time updates

What You Can Do with Tyk GraphQL:

- Securely expose existing GraphQL APIs
- Use Tyk's integrated engine to create a Universal Data Graph – combine multiple services into one unified GraphQL endpoint